

ProtoQSAR guide: reading CheMatSustain test data

Beginner-friendly instructions for test and experimental-data access

This guide is for ProtoQSAR users. It starts from the beginning and assumes you have not used an API before.

1. What you receive

Each ProtoQSAR user receives their own two credential values:

```
client_id:      cms_...  
client_secret: ...
```

These values are different from your website email and password.

- The **client ID** identifies your API credential.
- The **client secret** proves that you are allowed to use it.
- Treat the client secret like a password.
- Do not share credentials with another user. Every ProtoQSAR user has their own.
- Never put the secret in source code, Git, screenshots, email, or support tickets.

Your credential can read:

- the complete test catalogue;
- experimental data for every test.

It does not grant protocol or shared-file access.

2. The API address

The base address is:

```
https://database.eurskem.com/api/v1
```

An **endpoint** is a path added to this address. For example, the tests endpoint is:

```
https://database.eurskem.com/api/v1/tests
```

All examples below use HTTPS. Do not change it to HTTP.

3. Easiest first test: curl

curl is a small command-line tool for making web requests. It is already installed on macOS and most Linux systems. On current Windows versions, use it in PowerShell or Windows Terminal.

macOS or Linux

Open Terminal and enter these commands. Replace the example values with your own credential:

```
export CHEMAT_CLIENT_ID='cms_your_client_id_here'
export CHEMAT_CLIENT_SECRET='your_client_secret_here'
```

The single quotation marks are important. They prevent the shell from treating characters in the secret as commands.

Check your credential:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/portal/me'
```

Then request the first 25 tests:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
'https://database.eurskem.com/api/v1/tests?limit=25&offset=0'
```

You should receive JSON: structured text beginning with [for a list or { for one object. A successful request has HTTP status 200.

When finished, remove the values from the current terminal session:

```
unset CHEMAT_CLIENT_ID CHEMAT_CLIENT_SECRET
```

Windows PowerShell

Open PowerShell and enter:

```
$env:CHEMAT_CLIENT_ID = 'cms_your_client_id_here'
$env:CHEMAT_CLIENT_SECRET = 'your_client_secret_here'
```

Check your credential:

```
curl.exe --fail-with-body --user "$($env:CHEMAT_CLIENT_ID): $($env:CHEMAT_CLIENT_SECRET)" `
"https://database.eurskem.com/api/v1/portal/me"
```

Request the first 25 tests:

```
curl.exe --fail-with-body --user "$($env:CHEMAT_CLIENT_ID): $($env:CHEMAT_CLIENT_SECRET)" `
  "https://database.eurskem.com/api/v1/tests?limit=25&offset=0"
```

When finished:

```
Remove-Item Env:CHEMAT_CLIENT_ID
Remove-Item Env:CHEMAT_CLIENT_SECRET
```

4. Available methods

All ProtoQSAR requests use the HTTP method GET. GET means “read”; these requests do not create, update, or delete test records.

Method and endpoint	Purpose
GET /portal/me	Check that the credential works
GET /tests?limit=25&offset=0	Read one page of test summaries
GET /experimental-data/{test_id}	Read detailed data for one test

Replace {test_id} with the numeric id returned by /tests. For example:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" `
  'https://database.eurskem.com/api/v1/experimental-data/3'
```

5. Understanding the test response

GET /tests returns a JSON list. Each item can contain:

Field	Meaning
id	Numeric identifier used by the experimental-data endpoint
organisation_id	Owning organisation, if assigned; legacy records may be null
work_package_name	CheMatSustain work package
element_cms_id	Consortium element/reference identifier
test_name	Name or type of test
test_details	Summary/details of the test
final_results	Final result summary, if present
statistical_analysis	Statistical analysis, if present
created_at	Creation time
updated_at	Last update time

GET /experimental-data/{test_id} can additionally return `raw_data` and `processed_data`. A field may be null when no value has been entered.

6. Pagination: obtaining every test

The API returns at most 25 tests per request. This is called **pagination**.

- `limit=25` asks for 25 records.
- `offset=0` starts at the first record.
- The next page uses `offset=25`, then 50, 75, and so on.
- Stop when a page contains fewer than 25 records.

Example second page:

```
curl --fail-with-body --user "$CHEMAT_CLIENT_ID:$CHEMAT_CLIENT_SECRET" \
  'https://database.eurskem.com/api/v1/tests?limit=25&offset=25'
```

Do not set `limit` above 25; the API will reject it.

7. Python: download all tests

Install Python 3, then install the `requests` package once:

```
python3 -m pip install requests
```

Create a file named `download_tests.py` with this complete program:

```

import json
import os

import requests

BASE_URL = "https://database.eurskem.com/api/v1"
CLIENT_ID = os.environ["CHEMAT_CLIENT_ID"]
CLIENT_SECRET = os.environ["CHEMAT_CLIENT_SECRET"]

all_tests = []
offset = 0

while True:
    response = requests.get(
        f"{BASE_URL}/tests",
        params={"limit": 25, "offset": offset},
        auth=(CLIENT_ID, CLIENT_SECRET),
        timeout=30,
    )
    response.raise_for_status()
    page = response.json()
    all_tests.extend(page)

    if len(page) < 25:
        break
    offset += 25

with open("chematsustain_tests.json", "w", encoding="utf-8") as output:
    json.dump(all_tests, output, indent=2, ensure_ascii=False)

print(f"Saved {len(all_tests)} tests to chematsustain_tests.json")

```

Set the environment variables as shown in section 3, then run:

```
python3 download_tests.py
```

The program creates `chematsustain_tests.json` in the current directory. It does not write your secret into that output file.

8. Python: retrieve experimental data

This example downloads test 3. Change `TEST_ID` to the test you need:

```

import json
import os

import requests

TEST_ID = 3
response = requests.get(
    f"https://database.eurskem.com/api/v1/experimental-data/{TEST_ID}",
    auth=(os.environ["CHEMAT_CLIENT_ID"], os.environ["CHEMAT_CLIENT_SECRET"]),
    timeout=30,
)
response.raise_for_status()

with open(f"experimental_data_{TEST_ID}.json", "w", encoding="utf-8") as output:
    json.dump(response.json(), output, indent=2, ensure_ascii=False)

```

9. Optional: save basic test fields as CSV

After downloading `chematsustain_tests.json`, this program creates a spreadsheet- friendly CSV file:

```

import csv
import json

with open("chematsustain_tests.json", encoding="utf-8") as source:
    tests = json.load(source)

columns = [
    "id",
    "work_package_name",
    "element_cms_id",
    "test_name",
    "test_details",
    "final_results",
    "statistical_analysis",
    "created_at",
    "updated_at",
]

with open("chematsustain_tests.csv", "w", newline="", encoding="utf-8-sig") as output:
    writer = csv.DictWriter(output, fieldnames=columns, extrasaction="ignore")
    writer.writeheader()
    writer.writerows(tests)

```

Open the resulting `chematsustain_tests.csv` in Excel or LibreOffice Calc.

10. Postman (no programming)

1. Open Postman and choose **New** → **HTTP Request**.
2. Set the method to **GET**.
3. Enter: `https://database.eurskem.com/api/v1/tests?limit=25&offset=0`
4. Open the **Authorization** tab.
5. Choose **Basic Auth**.
6. Put your `client_id` in **Username**.
7. Put your `client_secret` in **Password**.
8. Click **Send**.
9. Confirm the response status is **200 OK**.

Avoid saving or synchronising a Postman collection containing the real secret. Prefer Postman's local/environment secret variables if you need to reuse it.

11. Common errors

Status or symptom	Meaning	What to do
200 OK	Request worked	Use the returned JSON
401 Unauthorized	Client ID/secret is wrong, disabled, or was copied with extra spaces	Copy both values again; if still failing, contact Eurskem
403 Forbidden	Credential lacks the required scope or resource access	Contact Eurskem; repeatedly retrying will not help
404 Not Found	Test ID/path does not exist, or is not visible	Check the ID came from <code>/tests</code> and check the URL
422 Unprocessable Entity	Invalid pagination, such as <code>limit</code> above 25 or negative <code>offset</code>	Use <code>limit=25</code> and an <code>offset</code> of 0 or more
500 Internal Server Error	Server-side problem	Record the time and endpoint, then contact Eurskem; do not send your secret
Empty list []	The request worked but returned no records on that page	If paginating, you have reached the end
Request times out	Network/VPN/firewall issue or temporary service issue	Retry once; then send the time and endpoint to support

When requesting support, provide:

- your work email;
- endpoint/path (without credentials);
- HTTP status;
- approximate time and timezone;

- a short description of what you expected.

Never include the client secret in a message or screenshot.

12. Security and credential rotation

- Store the secret in a password manager or approved secret manager.
- Use environment variables for scripts; do not paste the secret into Python.
- Add `.env` files to `.gitignore` before using them.
- Do not disable TLS certificate verification (`verify=False`, `-k`, or `--insecure`).
- Do not share one user's credential among colleagues.
- If a secret is lost or may have been exposed, stop using it and notify Eurskem immediately. Eurskem will disable it and issue a replacement; secrets cannot be recovered from the server.

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